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Savitribai Phule Pune University

Information Technology - Third Year

Computer Network and Security - Unit I

APPLICATION LAYER

Client Server Paradigm: Communication using TCP and UDP, Peer to Peer Paradigm, Application Layer Protocols: DNS, FTP, TFTP, HTTP, SMTP, POP, IMAP, MIME, DHCP, TELNET.

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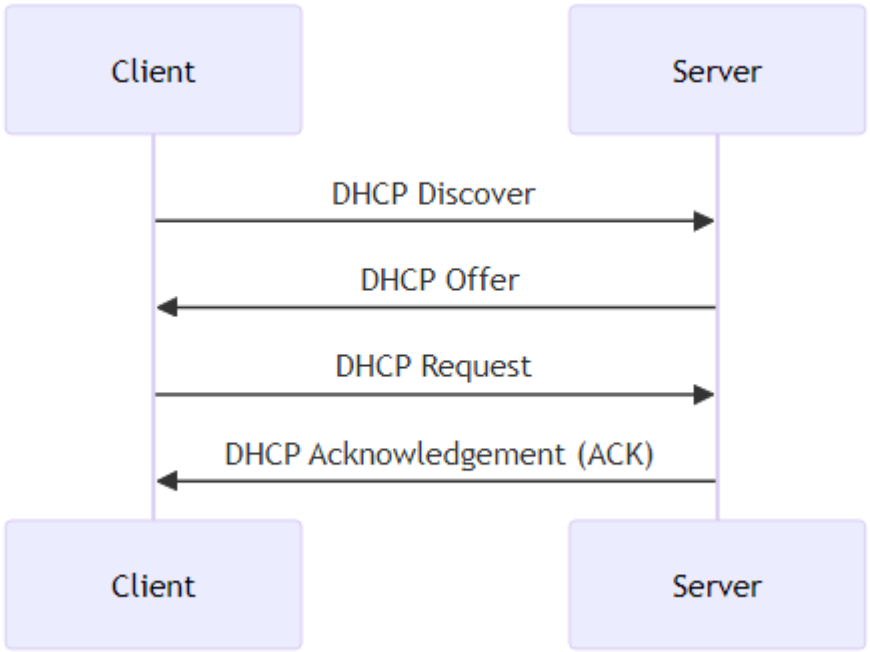
Q1. Explain Stepwise Procedure Carried out by DHCP Client and DHCP Server.

Definition

DHCP (Dynamic Host Configuration Protocol) is a network protocol used to automatically assign IP addresses and other network configuration parameters to devices in a network.

It works on client-server model.

Diagram



Caption: DORA process in DHCP communication.

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Stepwise Procedure (DORA Process)

DHCP process is called DORA:

1) DHCP Discover

- Client broadcasts a request
- Message is sent to find available DHCP servers
- Source IP = 0.0.0.0
- Destination IP = 255.255.255.255

2) DHCP Offer

- Server replies with available IP address
- Includes subnet mask, gateway, DNS, lease time

3) DHCP Request

- Client selects one offer
- Sends request to confirm selection

4) DHCP Acknowledgement (ACK)

- Server confirms assignment
- IP address is leased to client

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Additional Points

- Uses UDP protocol
- Port numbers:

Server → 67

Client → 68

- IP address is given for limited lease time

Conclusion

DHCP automatically assigns IP addresses using DORA process, reducing manual configuration.

Example

When you connect laptop to WiFi, DHCP automatically assigns IP address.

Q2. Define DNS. Explain Name Resolution and Resource Records.

Definition

DNS (Domain Name System) is a system that translates human-readable domain names into IP addresses.

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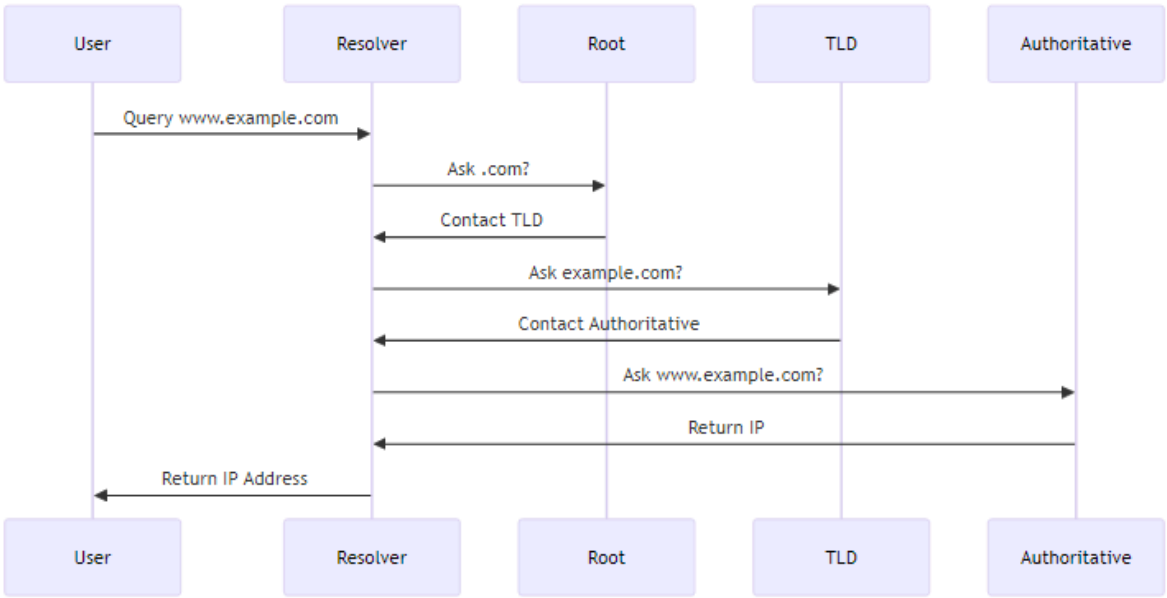
Example:

www.google.com → 142.250.183.206

How Name Resolution Happens

Diagram



Caption: DNS name resolution process.

Stepwise Explanation

1. User enters domain name
2. Resolver checks local cache
3. If not found, query sent to Root Server

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4. Root directs to TLD server (.com, .org)
5. TLD directs to Authoritative server
6. Authoritative server gives IP address
7. Resolver returns IP to user

DNS Resource Records and Functions

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Record	Full Form	Function
A	Address Record	Maps domain to IPv4 address
AAAA	IPv6 Address	Maps domain to IPv6 address
CNAME	Canonical Name	Alias of another domain
MX	Mail Exchange	Specifies mail server
NS	Name Server	Identifies authoritative server
SOA	Start of Authority	Contains domain admin info
PTR	Pointer Record	Reverse DNS lookup
TXT	Text Record	Stores text info (SPF etc.)
SRV	Service Record	Defines service location

Conclusion

DNS converts domain names into IP addresses using hierarchical distributed database system.

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Example

Typing www.amazon.com gives IP address of Amazon server.

Q3. What is Domain Name System Example? Explain Hierarchical Structure of DNS.

Example of DNS

If we type:

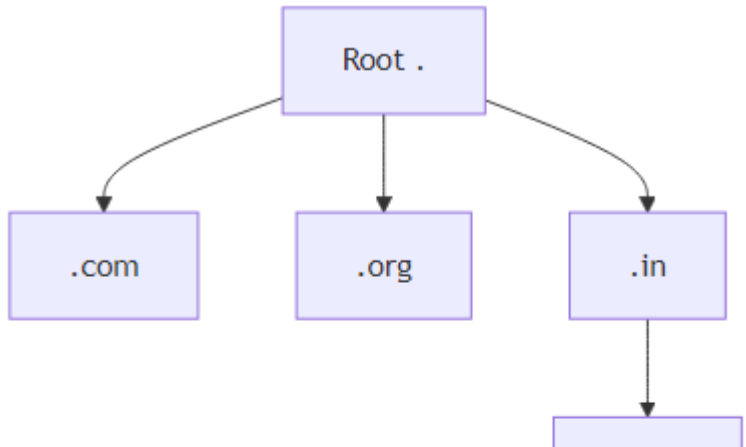
www.sppu.ac.in

DNS translates it into corresponding IP address.

Hierarchical Structure of DNS

DNS follows tree-like hierarchical structure.

Diagram



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Caption: Hierarchical structure of DNS.

Levels of DNS Hierarchy

1) Root Level

- Represented by dot (.)
- Top of hierarchy

2) Top Level Domain (TLD)

- .com
- .org
- .edu
- .in

3) Second Level Domain

- google.com
- sppu.ac.in

4) Subdomain

- www.google.com
- mail.google.com

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Explanation in Simple Words

DNS is organized like inverted tree:

Root → TLD → Domain → Subdomain

Each level is separated by dot (.).

Conclusion

DNS is hierarchical and distributed. It ensures scalable and efficient name resolution across the internet.

Example

Root → .com → google.com → www.google.com

Q4. What is MIME? Discuss its Role in SMTP.

Definition

MIME (Multipurpose Internet Mail Extensions) is an extension of SMTP that allows sending multimedia content like images, audio, video, and attachments through email.

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SMTP originally supports only 7-bit ASCII text.

MIME overcomes this limitation.

Diagram

Caption: MIME enables multimedia data transfer through SMTP.

Why MIME is Needed in SMTP?

SMTP limitations:

- Supports only plain text
- Cannot send binary files
- Limited character encoding

MIME solves this by:

- Converting binary data into ASCII format
- Supporting multiple content types
- Supporting attachments

Role of MIME in SMTP

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1. Converts non-ASCII data into ASCII format
2. Adds headers describing content type
3. Allows multiple parts in one message
4. Enables file attachments

Example

Sending a PDF file via email:

- MIME encodes PDF
- SMTP sends encoded text
- Receiver decodes it back to PDF

Conclusion

MIME enhances SMTP by enabling multimedia email communication.

Q5. What is MIME? Explain MIME Header in Detail. Discuss its Role in SMTP.

Definition

MIME is a protocol that extends email format to support multimedia and multiple content types.

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It adds additional headers to standard email message.

MIME Header Format

MIME header is added before the body of email.

Important MIME Headers

1) MIME-Version

Specifies MIME version used.

Example:

MIME-Version: 1.0

2) Content-Type

Specifies type of data in message.

General format:

Content-Type: type/subtype

Examples:

More Stuff Inside → Click here

Content-Type: text/plain

Content-Type: text/html

Content-Type: image/jpeg

Content-Type: application/pdf

3) Content-Transfer-Encoding

Specifies encoding method.

Examples:

- 7bit
- 8bit
- base64
- quoted-printable

Base64 is commonly used for attachments.

4) Content-Disposition

Specifies how content should be displayed.

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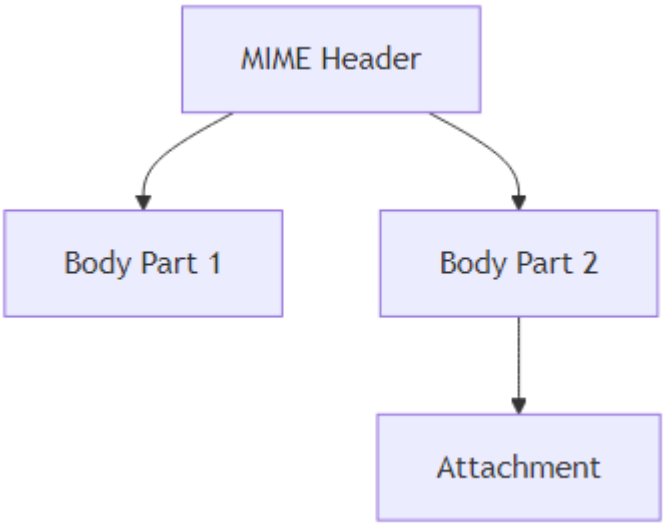
Example:

Content-Disposition: attachment; filename="file.pdf"

MIME Message Structure



Caption: Structure of MIME formatted email.

Role of MIME in SMTP

- SMTP sends only text
- MIME encodes multimedia into text
- SMTP transmits encoded message
- Receiver decodes using MIME headers

Thus MIME works on top of SMTP.

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Conclusion

MIME adds structured headers and encoding methods to allow SMTP to send multimedia emails and attachments.

Q6. Why FTP Requires Two TCP Ports? Explain Four Commands of FTP.

Definition

FTP (File Transfer Protocol) is used to transfer files between client and server over TCP/IP network.

Why FTP Uses Two TCP Ports?

FTP uses:

- Port 21 → Control connection
- Port 20 → Data connection

Reason

FTP separates:

1. Control connection → For commands
2. Data connection → For actual file transfer

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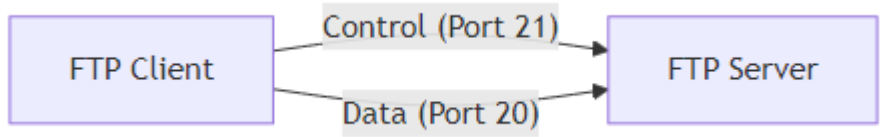
This improves efficiency.

Diagram



Caption: FTP uses separate ports for control and data.

Explanation

Control Connection

- Used for sending commands
- Remains open during session

Data Connection

- Used for transferring files
- Opened and closed for each file transfer

Four FTP Commands

1) USER

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Used to send username to server.

Example:

USER admin

2) PASS

Used to send password.

Example:

PASS 1234

3) LIST

Displays list of files in directory.

Example:

LIST

4) RETR

Used to download file from server.

More Stuff Inside → Click here

Example:

RETR file.txt

Additional Commands (for knowledge)

- STOR → Upload file
- QUIT → Close connection

Conclusion

FTP uses two TCP ports to separate control and data communication. Commands like USER, PASS, LIST, and RETR help in managing file transfer sessions.

Example

When downloading a file from FTP server:

- Control port handles login
- Data port transfers file

Q7. Compare FTP and TFTP. List 5 Commands of Each.

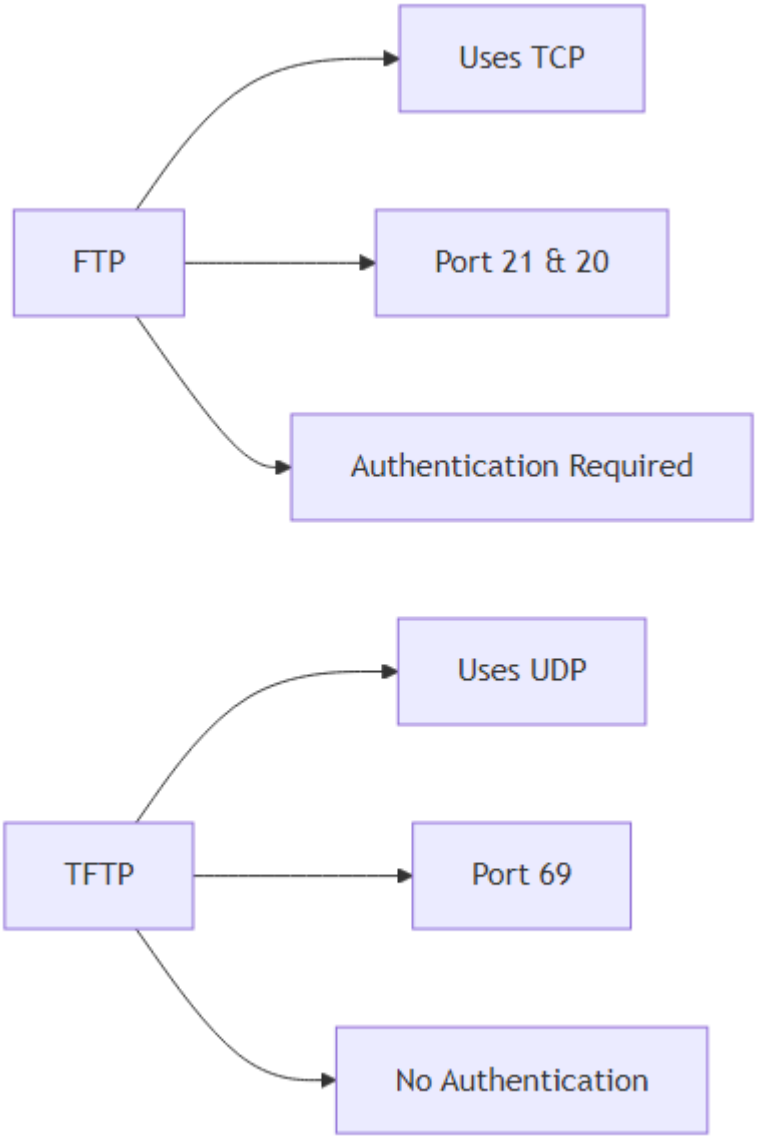
Definition

More Stuff Inside → Click here

FTP (File Transfer Protocol) is a reliable protocol used to transfer files between client and server over TCP.

TFTP (Trivial File Transfer Protocol) is a simple file transfer protocol used over UDP with limited features.

Diagram



More Stuff Inside → Click here

Caption: Basic difference between FTP and TFTP.

Difference Between FTP and TFTP

Basis	FTP	TFTP
Full Form	File Transfer Protocol	Trivial File Transfer Protocol
Transport Layer	TCP	UDP
Port Number	21 (control), 20 (data)	69
Authentication	Required	Not required
Reliability	Reliable	Less reliable
Features	Many commands	Very limited commands
Usage	Large file transfer	Boot files, firmware

5 Commands of FTP

More Stuff Inside → Click here

1. USER - To send username
2. PASS - To send password
3. LIST - To display file list
4. RETR - To download file
5. STOR - To upload file

5 Commands of TFTP

TFTP has limited operations (not command-based like FTP).

Common operations:

1. RRQ - Read Request (download file)
2. WRQ - Write Request (upload file)
3. DATA - Transfer data
4. ACK - Acknowledge packet
5. ERROR - Indicate error

Conclusion

FTP is feature-rich and reliable.

TFTP is simple and lightweight, mainly used for small transfers.

Q8. Differentiate between POP3 and IMAP.

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Definition

POP3 (Post Office Protocol version 3) and IMAP (Internet Message Access Protocol) are used to retrieve emails from mail server.

Difference Between POP3 and IMAP

Basis	POP3	IMAP
Full Form	Post Office Protocol v3	Internet Message Access Protocol
Port	110	143
Storage	Downloads and deletes email	Syncs with server
Access	Single device	Multiple devices
Folder Support	No	Yes
Synchronizati	Not supported	Supported

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Explanation

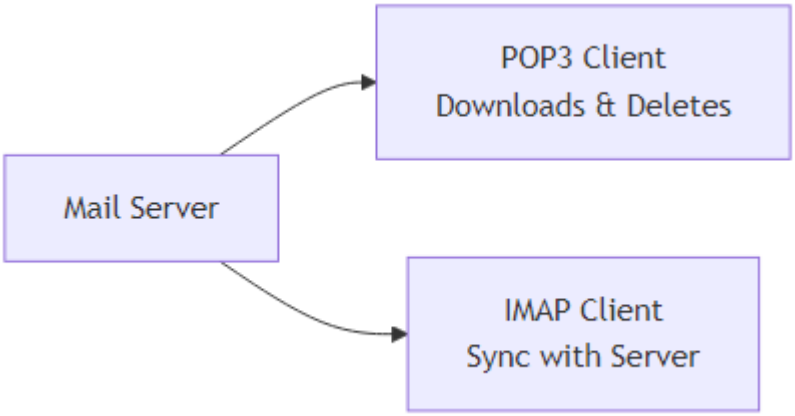
POP3

- Downloads emails to local device
- Usually deletes from server
- Good for single-device usage

IMAP

- Keeps emails on server
- Allows multiple device access
- Synchronizes changes

Diagram



Caption: POP3 vs IMAP working.

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Conclusion

POP3 is simple and device-based.

IMAP is advanced and supports synchronization across devices.

Q9. Explain Role of SMTP and POP Protocol in E-mail Message Transfer.

Definition

SMTP (Simple Mail Transfer Protocol) is used to send email.

POP (Post Office Protocol) is used to receive email.

Role of SMTP

- Transfers mail from sender to mail server
- Transfers mail between mail servers
- Uses TCP port 25

Role of POP

- Retrieves mail from mail server
- Allows user to read messages

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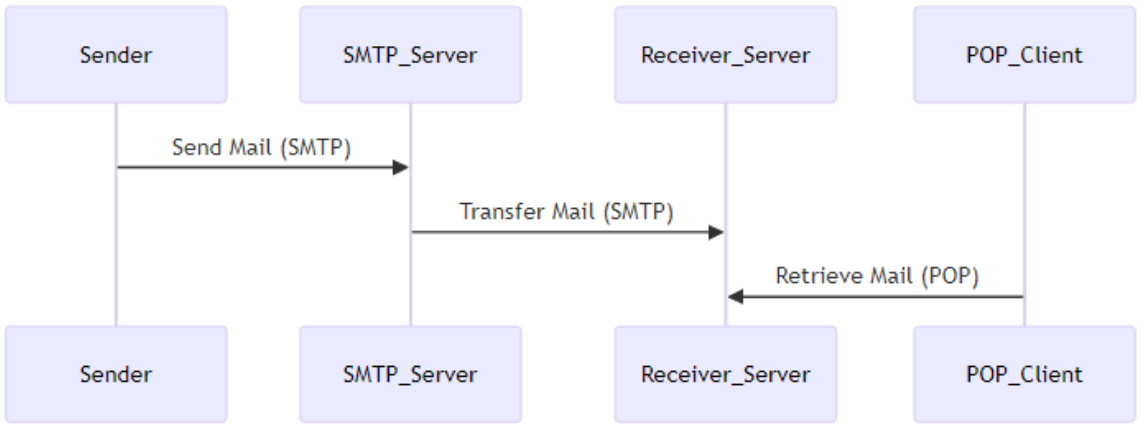
- Uses TCP port 110

Diagram



Caption: Email transfer using SMTP and POP.

Stepwise Working

1. User composes email
2. SMTP sends mail to sender mail server
3. SMTP transfers mail to receiver mail server
4. POP retrieves mail to receiver device

Conclusion

SMTP is used for sending emails.

POP is used for receiving emails.

More Stuff Inside → Click here

Together they complete email communication process.

More Stuff Inside → Click here